Purushothaman Natarajan

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Summary

Senior AI Systems Engineer with 5+ years architecting and owning production-grade GenAI, Agentic AI, and Computer Vision platforms across enterprise domains. Specialized in designing hybrid RAG systems, multi-agent orchestration pipelines, and cost-governed LLM deployments — with measurable outcomes including 91% reduction in manual processing load and 85% reduction in document review effort across high-volume enterprise workflows. Brings end-to-end ownership across system design, retrieval strategy, prompt optimization, cost governance, and post-deployment observability. DRDO-funded researcher with three peer-reviewed publications and open-source contributions. Experienced in stakeholder alignment, async cross-functional delivery, and translating ambiguous business requirements into robust, monitored production systems.

Skills

LLM Systems & GenAI: GPT-4, Claude, LLaMA, Mistral, Qwen; Prompt Engineering, DSPy Optimization, LoRA Fine-tuning, Instruction Tuning, Rule-based & Character-level Guardrails (enterprise), Tool Calling, Model Routing, Token Budget Governance
Agentic AI & Orchestration: LangGraph (stateful deterministic workflows), LangChain, DSPy (prompt optimization), AutoGen, CrewAI; Multi-Agent Design, A2A Communication, Tool Orchestration
RAG & Retrieval: Hybrid BM25 + Dense Retrieval, Cross-Encoder Reranking, Semantic/Recursive Chunking, FAISS, Pinecone, Milvus, USearch; RAGAS, TruLens Evaluation
Observability & MLOps: Langfuse, MLflow, Arize Phoenix; CI/CD Pipelines, Prompt Regression Testing, Model Drift Monitoring, Latency Dashboards, Docker, Kubernetes, FastAPI
Computer Vision & Document AI: YOLO, SAM, SegFormer, DeepLabV3+, ViT-STR, MGP-STR, SwinIR, BSRGAN; Object Detection, Segmentation, OCR, Edge Inference, ONNX
ML & Deep Learning: Transformers, CNNs, RNNs, LSTM, Diffusion Models, GANs, SimCLR, RL (PPO, DPO), LIME, SHAP, Grad-CAM
Cloud & Infrastructure: Azure (AI Foundry, ML), GCP
NLP: NER, Text Classification, Information Retrieval, Topic Modeling, PII Detection
Databases: MySQL, PostgreSQL, Neo4j, LMDB, Vector DBs
Programming: Python (primary), SQL, C++ (basic), JavaScript (basic)
Security & AI Risk: CVE Analysis, Threat Intelligence Ingestion, Risk Scoring, Secure ML Deployment

Experience

Senior Forward Deployed Engineer, UiPath – Bangalore, IN Sept 2026 – Present

- Recently joined UiPath as Senior Forward Deployed Engineer. *(Details to be added as scope and deliverables are finalized.)*

Data Scientist (AI/ML & Agentic Systems), Blackstraw AI – Chennai, IN Nov 2024 – Aug 2026

- Architected a production-grade hybrid RAG + multi-agent platform spanning document intelligence, vulnerability research, and product data pipelines — owning system design, retrieval strategy, model selection, and deployment — reducing a team of 5's manual processing load from ∼110 hrs/week to under 10 hrs/week (91% reduction across 15K+ monthly document queries).
- Fine-tuned Qwen-72B for item coding classification via LoRA instruction tuning on domain-specific classification guidelines — replacing the DSPy prompt optimization approach and achieving **50–60% workflow automation** at >**95% accuracy** across 3,600+ item categories; selected LangGraph for stateful deterministic orchestration of the fine-tuned model's multi-step decision pipeline.
- Designed recursive + semantic chunking strategy optimized for long enterprise PDFs; benchmarked OpenAI, BGE, and E5 embedding models for domain fit; implemented hybrid BM25 + dense retrieval with cross-encoder reranking; evaluated pipeline quality via RAGAS on a curated 300-sample golden set (faithfulness, answer relevancy, context precision).
- Reduced LLM inference and search costs by ∼35% through prompt compression, aggressive response caching,

and a model routing layer dynamically switching between GPT-4 and smaller SLM fallbacks based on task complexity; introduced per-workflow token budget governance.

- Built production observability stack using Langfuse trace logging and custom evaluation harnesses — monitoring agent tool call reliability, hallucination rates, latency distributions, and prompt regression across deployed agentic systems.
- Architected OCR and document intelligence stack (SegFormer, DeepLabV3+, ViT-STR, SwinIR/BSRGAN super-resolution) — cutting manual review from ∼8 hrs/day across 1,000+ daily documents to under 1.5 hrs (∼85% reduction); improved OCR extraction F1 from 74% to 91% on complex scanned receipts.
- Built CI/CD pipelines for model versioning and rollback; implemented prompt regression testing and experiment tracking via MLflow; monitored model drift and inference latency across production deployments.
- Implemented automated PII detection and redaction pipelines across documents and images using NLP + embedding similarity matching.
- Designed multi-agent CrewAI workflows for automated codebase migration (Oracle→MySQL, Java→Python) with validation and tool orchestration stages.

Research Fellow (Machine Learning & eXplainable AI), SRMIST – Chennai, IN Sept 2023 – Oct 2024

- Selected as DRDO-funded PhD research fellow focused on explainable and reliable AI systems for defense-grade sonar imaging applications.
- Developed deep learning models for underwater sonar object detection and classification under low-signal and noisy sensing conditions.
- Built explainability-first ML pipelines integrating LIME, SHAP, and Grad-CAM to produce interpretable predictions for mission-critical AI systems.
- Delivered self-explainable AI models validated and tested by NPOL and DRDO teams for operational underwater surveillance scenarios.

Machine Learning Associate, Amazon – Chennai, IN Jul 2022 – May 2023

- Improved annotation and data quality pipelines for Alexa, Ring, and Halo ML systems using NLP preprocessing and normalization workflows.
- Standardized labeling and validation processes, reducing downstream model noise and improving training dataset consistency.
- Built performance dashboards and QA metrics tracking that enabled faster feedback cycles and process optimization.

Customer Support Executive, Amazon – Coimbatore, IN Aug 2021 – Nov 2021

- Managed customer and seller queries, refunds, and returns, achieving top performer status for two consecutive months.

Education

SRM University, PhD in Computer Science [**Drop-out**] Feb 2024 – Nov 2024

BITS, Pilani, M.Tech in Data Science Sept 2022 – Sept 2024

Anna University, B.E in Mechanical Engineering Aug 2015 – Nov 2020

Publications

Synth-SONAR: Sonar Image Synthesis with Enhanced Diversity and Realism via Dual Diffusion Models and GPT Prompting Sept 2024 *Cited by 3*

Purushothaman Natarajan, Kamal Basha, Athira Nambiar

VALE: A Multimodal Visual and Language Explanation Framework for Image Classifiers using eXplainable AI and Language Models June 2024 *Cited by 10*

Purushothaman Natarajan, Athira Nambiar

Underwater SONAR Image Classification and Analysis using LIME-based Explainable Artificial Intelligence Dec 2023 *Cited by 5*

Purushothaman Natarajan, Athira Nambiar

Projects

Autonomous Vulnerability Intelligence Platform Enterprise

- Designed and owned a production autonomous vulnerability intelligence platform — architecting the ingestion, reasoning, and remediation layers end-to-end across continuously updated CVE and security intelligence sources.
- Implemented hybrid BM25 + dense retrieval with cross-encoder reranking for precise CVE context retrieval; selected BGE-large as embedding backbone after benchmarking against OpenAI and E5 across domain-specific recall; evaluated pipeline using RAGAS on a 300-sample golden set.
- Built hierarchical multi-agent orchestration: triage agent → exploitability reasoning agent → remediation packaging agent, with state persistence across LangGraph workflow nodes.
- Introduced token budget governance and GPT-4/SLM routing to reduce per-query inference cost by ∼30% while maintaining output quality on high-priority CVEs.
- Monitored agent reliability in production via Langfuse trace logging — tracking tool call success rates, reasoning consistency, and hallucination frequency across live CVE ingestion cycles.
- Engineered unhappy-path resilience: implemented per-node retry logic with exponential backoff, fallback chains routing to simpler SLMs on tool call failure, and graceful degradation modes that return partial structured output rather than silent failure — reducing pipeline crash rate from ∼18% to under 3% in production.
- **Tools:** Python, Azure workflows, LLM Agents, AI Foundry, Hybrid RAG, Vector DB, RAGAS, Langfuse, Workflow Orchestration.

Retail Vision Intelligence — Detection & Large-Scale Classification on Edge Devices Enterprise

- Architected edge-deployable computer vision system for retail shelf monitoring — owning model selection, inference optimization, and deployment pipeline across complex in-store environments.
- Designed a vector search based classification system supporting **200K+ product labels** — chosen over traditional softmax classification for scalability and zero-shot extensibility; improved top-5 retrieval accuracy by 14% over baseline using SimCLR-pretrained embeddings.
- Optimized inference pipeline latency from ∼340ms to under 80ms on edge devices through model quantization and ONNX export, enabling real-time shelf analytics.
- **Tools:** YOLO, Image Embeddings, Vector Search, OpenCV, PyTorch, ONNX, Edge AI Optimization.

Hierarchical LLM Decision Automation Engine for Item Coding Enterprise

- Designed a hierarchical LLM reasoning engine to replace manual item coding workflows across 3,600+ categories — architecting the playbook generation, decision tree structuring, and validation layers end-to-end.
- Built image-text noise filtering and rule-based validation layers improving pipeline robustness across noisy product metadata inputs.
- **Tools:** DSPy, Python, LLMs, Hierarchical Reasoning, Image-Text Processing, Rule-based Validation.

Agentic Product Packaging Intelligence System Enterprise

- Architected a search-augmented LLM classification system to automate standards-based product packaging code mapping — replacing a fully manual analyst workflow across thousands of SKUs.
- Designed a multi-stage pipeline: SERP retrieval → attribute extraction → DSPy-optimized classification → rule-based character-level guardrail validation, ensuring output codes conformed to enterprise packaging standards before write-back.
- Reduced manual packaging code lookup and enrichment time from ∼6 min/SKU to under 45 seconds, enabling processing of 500+ SKUs/day with a 2-person team.
- Built confidence-scored output layer routing low-certainty predictions to human review, maintaining >95% precision on auto-approved codes.
- **Tools:** DSPy, Python, SERP API, LLM Reasoning, NLP Pipelines, Rule-based Validation.

Logistics Asset Detection & Cross-Location Package Similarity Tracking Enterprise

- Built computer vision pipelines to detect logistics assets including packages, cardboard boxes, pallets, and forklifts across warehouse environments.
- Implemented embedding-based similarity matching to determine whether packages observed across different warehouse locations correspond to the same physical item.
- Improved clustering quality and reduced misclassification using SimCLR-based representation learning and anomaly detection methods.
- **Tools:** YOLO, SimCLR, CNNs, Embedding Similarity Search, OpenCV, PyTorch.

Enterprise Document Intelligence Platform — OCR Receipts Enterprise

- Architected a production document AI pipeline converting large-scale scanned receipt and retail documents into structured analytical datasets — owning model selection, preprocessing stack, and post-processing validation.
- Improved OCR extraction F1 from 74% to 91% on a 2,000-document held-out test set of complex multi-column retail receipts, using FFT-former deblurring + SwinIR/BSRGAN super-resolution as preprocessing stages ahead of ViT-STR.
- Reduced super-resolution inference latency from ∼16s to 2–3s through INT8 quantization of the SwinIR model and tile-based parallel batch processing — without measurable degradation in downstream OCR quality.
- Built layout-aware post-processing using reading-order detection and structured field extraction, reducing downstream data cleaning effort by ∼60%.
- **Tools:** ViT-STR, SwinIR, BSRGAN, FFT-former, ONNX, PyTorch, NLP, Data Post-processing.

Agentic Item Description Expansion using LangChain & LangGraph Enterprise

- Built a multi-agent pipeline to expand abbreviated product titles into rich, structured item descriptions for large-scale e-commerce catalogs.
- Designed LangGraph-based multi-step workflows integrating retrieval, reasoning, and intelligent generation; used semantic chunking with sliding-window overlap and embedding-based reranking to improve retrieved context relevance.
- Implemented LangChain tool-calling agents for product metadata retrieval, attribute extraction, and validation.
- **Tools:** LangChain, LangGraph, LLM Agents, lmdb, Python, Prompt Engineering, Retrieval Pipelines.

DL-Studio — Unified AI Deep Learning Lab Open Source

- Unified deep learning experimentation platform for regression analysis, automated data preprocessing, and competitive benchmarking across 14+ traditional ML and deep learning architectures.
- Repo: github.com/Purushothaman-natarajan/doctane
- **Tools:** Python, PyTorch, Scikit-learn, Streamlit.

Doctane — Modular OCR & Document AI Framework Open Source

- Built an open-source OCR and document understanding repository integrating state-of-the-art models.
- Implemented modular pipelines for text extraction, preprocessing, and document intelligence tasks.
- Repo: github.com/Purushothaman-natarajan/doctane
- **Tools:** Python, Transformers, PyTorch, NLTK, Streamlit.

Explainable Remote Sensing Vision Framework Research

- Built a transfer learning-based remote sensing scene classifier with model explainability.
- Applied LIME and Grad-CAM to generate visual explanations for model predictions.
- Repo: github.com/Purushothaman-natarajan/eXplainable-AI-for-Image-Classification-on-Remote-Sensing
- **Tools:** Python, TensorFlow, Scikit-learn, LIME, Grad-CAM, Gradio.

Additional Experience and Awards

Independent AI Systems Consultant (Upwork & LinkedIn, 2022–2024): Independently designed and delivered end-to-end AI and ML systems for international clients across NLP, computer vision, and data pipeline domains — projects valued between $10K–$20K. Demonstrates that production delivery depth transfers beyond a single employer context.

Instructor, BrightNext Academy (2023–2024): Taught Machine Learning and Deep Learning courses to over 100 students.

Business Analyst, IIFL (Oct 2019 – May 2021): Managed daily client transactions worth $10M, liaised between clients and the trading desk, and provided strategic market insights.

Third Prize, Innovation & Design on Remote Sensing Data by Hack2Skill: Recognized for designing a synchronized research and production-ready dashboard for an explainable image classifier, competing against 100+ entries.

Licenses & Certifications

Udacity: Deep Learning, Computer Vision, Generative AI Nanodegrees.

DeepLearning.AI: Retrieval Augmented Generation (RAG), Reliable AI using Guardrails.

Codecademy: BI Dashboards with Power BI & Tableau, Business Intelligence Data Analyst, SQL for Marketers and Product Managers.

LinkedIn: Advanced SQL, AI Foundations (ANN, CNN, R-CNN, RNN, LSTM, GNN & Transformers), Advanced NLP with Python, GANs and Diffusion Models with TensorFlow and PyTorch, Transfer Learning Using PyTorch, Deep Learning for Computer Vision Applications.